THRONG
EVOLUTION

Overview

Throng: Evolution is a single-player simulation/strategy game focused on guiding a population of creatures through the pressures of survival and evolution. The player's role can be described as part observer, part "nature custodian." Rather than directly controlling individual creatures, the player influences the environment and initial conditions, then watches natural selection play out and can intervene strategically. The experience is somewhat akin to being an ecologist running an experiment or a deity overseeing a small ecosystem.

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Target Audience

Middle to high school biology students, and the general public interested in science education.Learning Outcomes:- Understand the mechanism of natural selection and how environmental pressures affect which individuals survive and reproduce.- Recognize that "survival of the fittest" refers to being best suited to an environment – not merely the biggest or strongest – and that even less "strong" individuals can survive if they are fit in other ways[1].- Appreciate how population dynamics (e.g. limited resources and carrying capacity) influence evolution, illustrating why not all offspring survive and why populations don't grow indefinitely[2].

Background & Research

Natural selection is a cornerstone of biology, yet numerous studies show it is widely misunderstood[3]. Many students – even some with biology education – struggle to grasp how evolution by natural selection truly works[3].

Misconceptions are the norm, and a solid understanding of natural selection is "the rare exception"[4]. One prevalent misunderstanding is the idea that "fittest" means the strongest or fastest individual. In reality, evolutionary "fitness" means being best suited to survive and reproduce in a given environment. As science educators note, "the fittest really is the 'good enough.' As long as you get a chance to reproduce and pass on your genes, you win."[1].

In other words, an organism that survives long enough to reproduce is "fit," even if it isn't the biggest or strongest. This serious game directly addresses that misconception by demonstrating scenarios where, for example, smaller or slower creatures might survive better if they require fewer resources or blend into the environment more effectively.

Fundamentally, Darwin's theory of natural selection was built on a few key observations about populations. Organisms have the potential to reproduce exponentially, leading to huge population growth in principle[5]. However, in nature we do not see unlimited growth; populations tend to stabilize because resources (food, space, etc.) are limited. Darwin described how most offspring do not survive to reproduce, due to starvation, predation, disease, and competition – thus population sizes remain roughly stable over time[2]. This imbalance between the capacity for exponential growth and the reality of limited resources creates a "struggle for existence"[6]. In Throng: Evolution, this principle is built into the simulation: creatures produce offspring, but only a fraction can survive given the food available, illustrating concepts of carrying capacity and population dynamics. By including these elements, the game connects ecological constraints to evolutionary outcomes, reinforcing why survival of the fittest emerges naturally from competition over resources.

Educational research suggests that active learning and simulations can improve understanding of complex scientific concepts. Traditional teaching about evolution often struggles to correct ingrained misconceptions, but learning-by-doing offers promise. Hands-on games and simulations have been used to demonstrate natural selection in classrooms with success[7]. For example, simple analog games (like card or board games) have helped students experience how selection works by "breeding" or removing individuals based on traits[7]. In a more technology-driven approach, interactive simulations allow learners to manipulate variables and see evolutionary processes unfold in real-time. One project by Cignoni et al. (2015) developed a digital predator–prey ecosystem where "the game lets the player modify the fundamental parameters of the species and observe the evolution of the ecosystem." The player could even intervene to add or remove individuals, experimenting with how the system balances out[8]. This kind of inquiry-based exploration aligns with modern educational strategies that emphasize learning through experimentation and observation.

Building on these insights, Throng: Evolution is designed as a serious game – a game with an explicit educational purpose. Unlike entertainment-only games, serious games are defined by their learning objectives (in this case, teaching natural selection)[9]. Research shows that well-designed serious games can indeed enhance student engagement, motivation, and even behavior toward learning[9]. By immersing players in a dynamic model of evolution, the game aims to leverage curiosity and interactivity to deepen understanding. Moreover, because players can directly see the consequences of changes (like a shift in available food or a mutation in traits), they can correct their misconceptions through visual feedback and experimentation.

Educational Rationale

Using a game-based approach for teaching evolution offers several potential advantages. Engagement and Motivation: Students are typically more engaged when learning feels like playing. Studies have found that gamified lessons and serious games significantly boost student motivation[10]. In the context of evolution (often considered a dry or abstract topic by learners), a game can spark interest and encourage students to spend more time thinking about evolutionary processes. The interactive nature of Throng: Evolution means learners are not just reading about natural selection – they are actively doing it, guiding a population through challenges. This active involvement can lead to better retention of concepts, as players form personal experiences with the material.

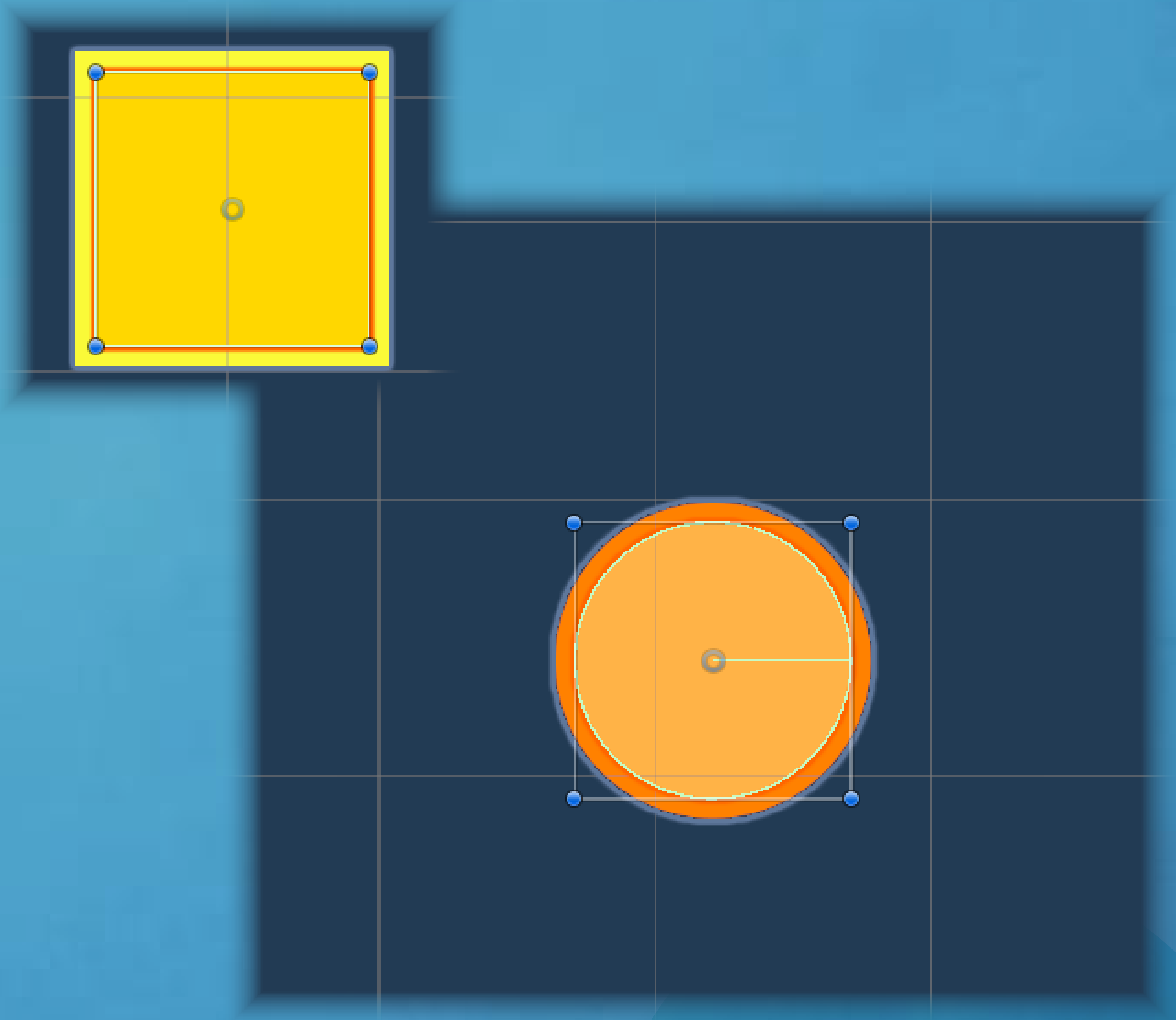
Visualization of Abstract Concepts: Evolution can be hard to grasp because it occurs over long timescales and involves statistical outcomes. A simulation speed s up time and lets players visualize abstract concepts like genetic variation, differential survival, and population change. For instance, rather than merely hearing that variation in a trait can confer a survival advantage, players will see that advantage play out (e.g. creatures with a certain trait surviving a famine). This aligns with the pedagogical idea that concrete experiences can solidify understanding of theoretical concepts.

Addressing Misconceptions through Exploration: Misconceptions such as "strength equals fitness" or "individuals evolve on purpose" can be corrected by letting players test and observe. In the game, if a player assumes only the fastest creatures will thrive, they might be surprised to find that in a scarce food scenario, the fastest (but high-metabolism) creatures actually starve, whereas slower, low-metabolism ones survive longer. Such outcomes directly confront prior assumptions, providing teachable moments. The game's design includes on-screen feedback cues and explanations (described later) to explicitly point out these outcomes, helping players connect the simulation events to real-world evolutionary principles.

Educational Rationale

Literature-Informed Design: Our approach is informed by findings in educational research. Serious games can improve learning experiences, but they must be carefully designed to maintain focus on learning goals. One concern researchers note is that students may get “overly focused on playing the educational games rather than learning from them.”[11] In light of this, Throng: Evolution incorporates features to keep the educational content front-and-center. For example, the game provides real-time data displays and “teaching tips” pop-ups that interpret what is happening in educational terms (such as noting when the population reaches a stable size, linking it to carrying capacity). By explicitly connecting gameplay events to scientific terminology and concepts, we aim to prevent the game from being a mindless exercise. Additionally, while games can boost motivation, some studies have shown mixed results on whether they boost achievement (knowledge gains) as much as traditional teaching[10]. To maximize learning, Throng: Evolution could be used alongside reflection activities: for instance, students might be prompted to predict outcomes, explain results, or answer quiz questions after gameplay. This blended approach leverages the engagement of the game and the rigor of traditional assessment to ensure both motivation and learning outcomes are high[10].

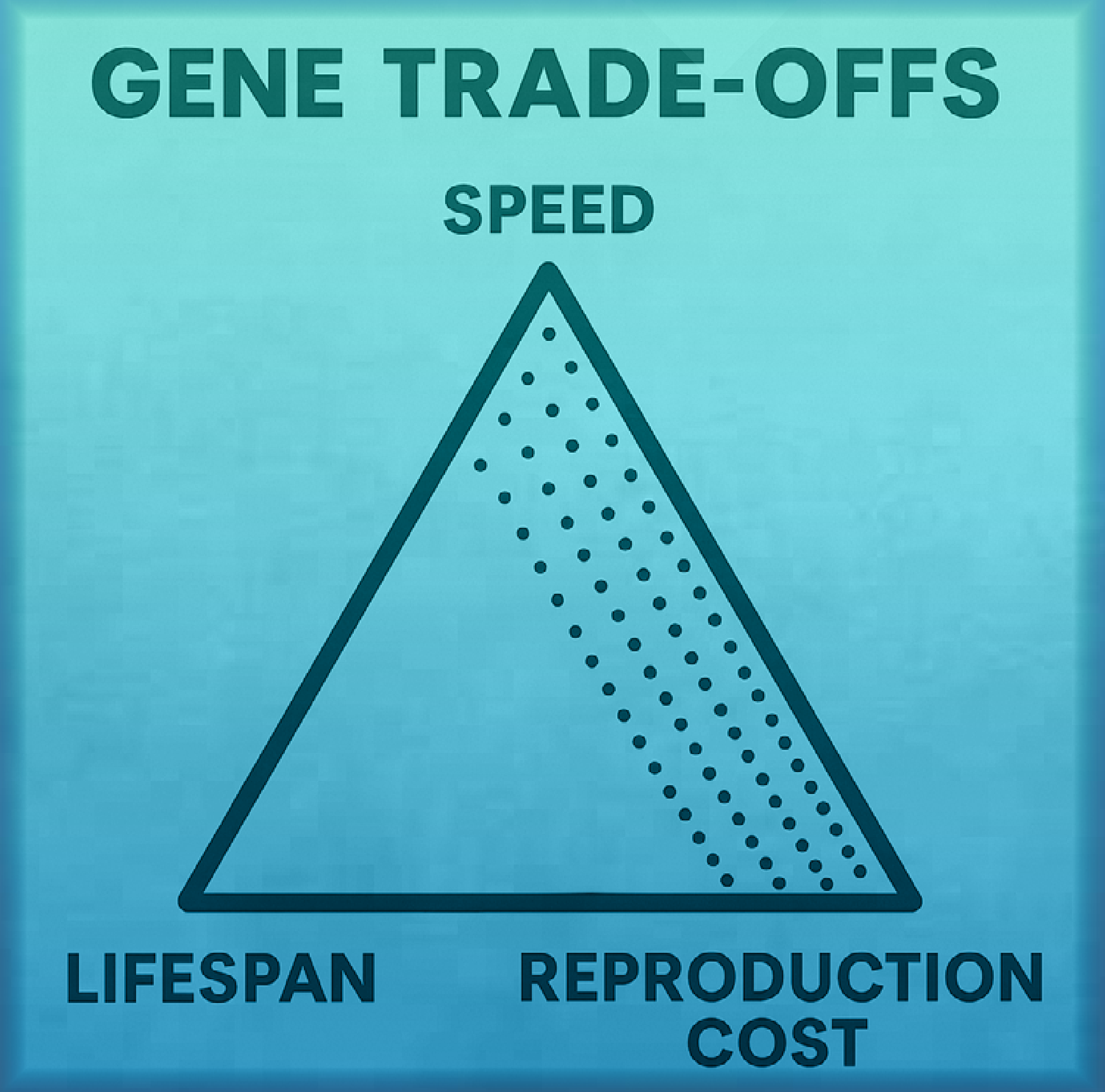
In summary, the rationale for Throng: Evolution is to provide an engaging, research-backed learning tool that makes the principles of natural selection intuitive. By simulating the evolutionary “game of survival,” students and players can learn-by-doing in a way that textbooks or lectures cannot easily replicate. The following sections detail the design of the game, including its core mechanics and how they map to educational concepts.



Game Concept

Throng: Evolution is a single-player simulation/strategy game focused on guiding a population of creatures through the pressures of survival and evolution. The player’s role can be described as part observer, part “nature custodian.” Rather than directly controlling individual creatures, the player influences the environment and initial conditions, then watches natural selection play out and can intervene strategically. The experience is somewhat akin to being an ecologist running an experiment or a deity overseeing a small ecosystem.

Game World & Scenario: The game is set in a simplified ecosystem represented in 2D. The core scenario centers on a species of small creatures (the “thronglets”) that must gather food, survive, and reproduce. There are no humanoid characters – the focus is on the creatures (and possibly predators or other factors in extended scenarios). Each round or session of the game represents several generations of these creatures. The environment contains resource patches (food sources) that the creatures need to consume to gain energy. Over time, resources can be depleted and regrow, and the environment can change (through player actions.). The visual style is clear and functional – creatures might be represented by simple shapes or icons with different colors or sizes indicating their traits, and resource patches are identifiable spots on the map.



Gameplay Loop

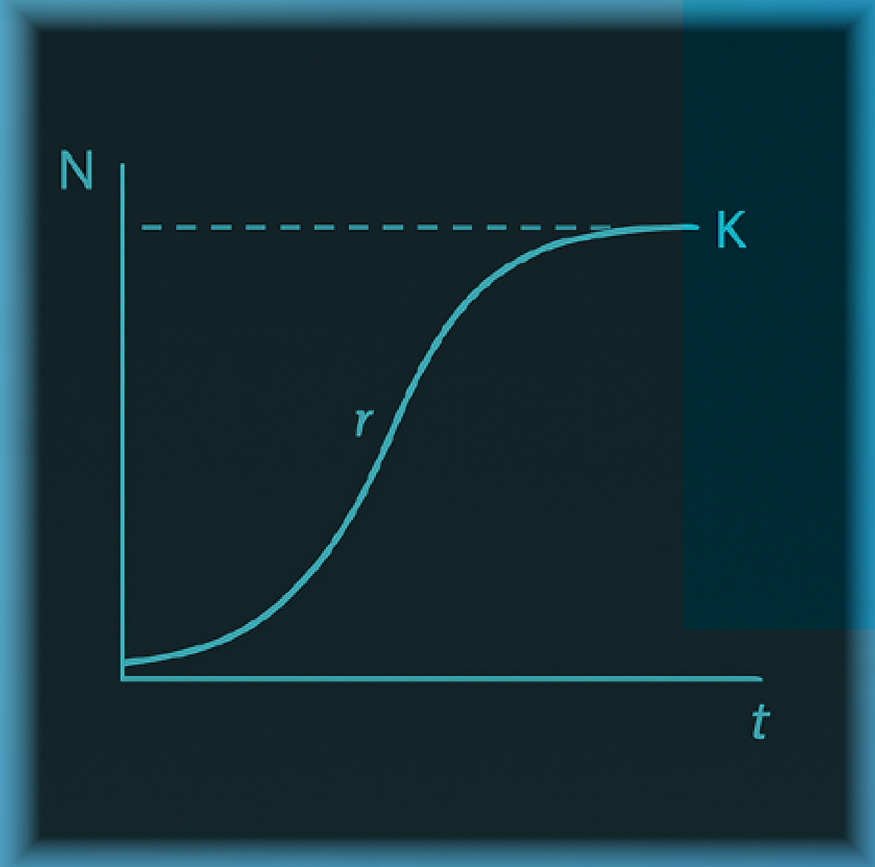
The high-level loop of Throng: Evolution

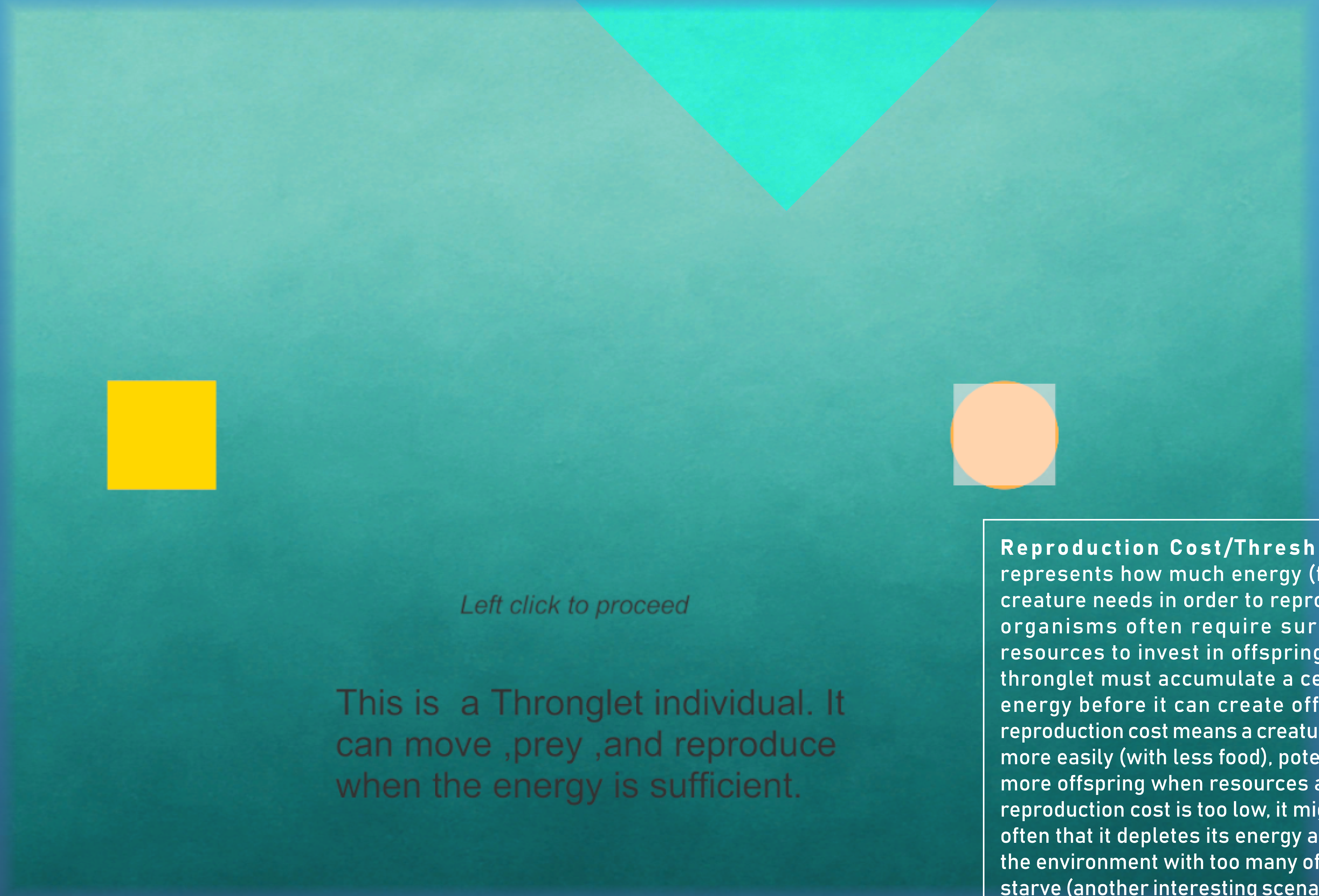
1. Set up conditions
2. Run simulation (creatures live, breed, die)
3. Observe outcomes
4. (Optional) intervene or adjust
5. Next generation evolves
6. Outcome/feedback

During the simulation phase, creatures autonomously move, eat, and reproduce according to the built-in rules of biology (described in Mechanics below). The player can adjust some parameters either at the start or during the run (depending on the mode) – for instance, introducing more food, toggling an environmental factor – to see how it affects the population. When the throng meets the requirement of the win-condition, the simulation stops and the player wins the game.

Gameplay & Mechanics

The game world provides the environmental context in which evolution occurs. It consists of a bounded area that contains resource patches (food sources). Each resource patch has a certain amount of food that regenerates over time. The total amount of available food effectively creates a carrying capacity for the creature population – there is only so much energy to go around. If the population grows too large for the environment to support, individuals will start to starve. This models the classic population dynamics concept: resources limit population growth[2].





The creatures (let's call them "Thronglets") are the agents of evolution in the game. Each thronglet has a set of heritable traits (genes) that affect its survival and reproductive success. In the current design, there are five key traits, each corresponding to a gene:

Lifespan: How long the creature can live (in simulation time) if it doesn't die from other causes. A longer lifespan increases the chance to find food and reproduce over the creature's life, but long-lived individuals might also consume more total resources. Shorter-lived individuals might reproduce earlier (this trait may co-vary with Reproduction cost).

Reproduction Cost/Threshold: This trait represents how much energy (food stored) the creature needs in order to reproduce. In nature, organisms often require surplus energy or resources to invest in offspring. In the game, a thronglet must accumulate a certain amount of energy before it can create offspring. A lower reproduction cost means a creature can reproduce more easily (with less food), potentially leading to more offspring when resources are limited. But if reproduction cost is too low, it might reproduce so often that it depletes its energy and dies or floods the environment with too many offspring that then starve (another interesting scenario to observe!).

Speed: Determines how fast the creature can move around the environment. A higher speed helps a thronglet reach food patches more quickly, which is advantageous if food is scarce or competition is high. However, speed comes at a cost (see Metabolism below).

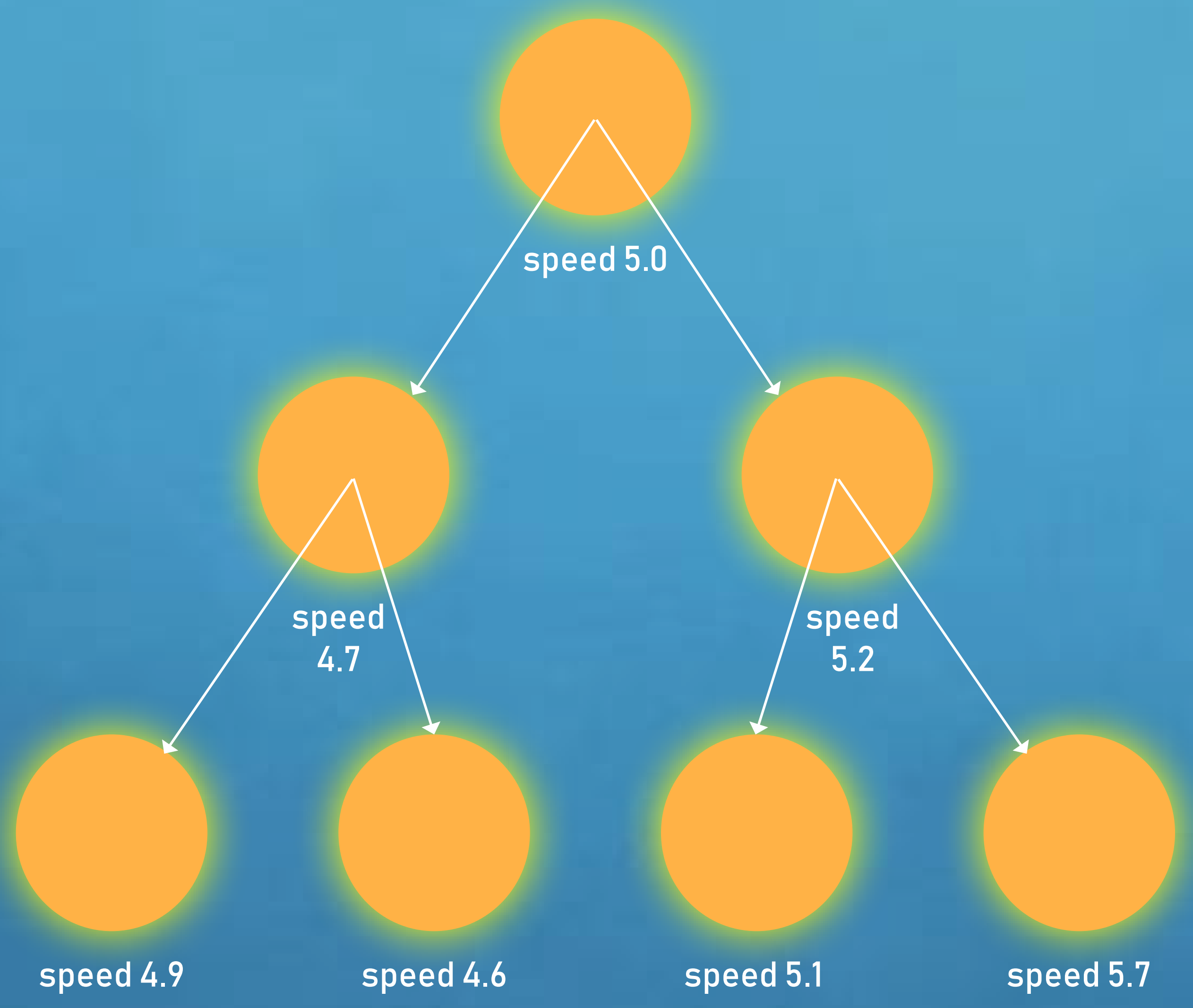
Metabolic Rate: This governs how fast the creature uses up energy (food). A higher metabolism means the creature must eat more frequently to avoid starvation. There is an implicit trade-off between Speed and Metabolism in our design – typically, creatures that move faster also consume energy faster. Thus, being swift can be a liability if food is sparse, reflecting real-life trade-offs (e.g. a shrew must eat almost constantly due to its high metabolism).

Gathering Efficiency: How effectively the creature harvests food from a patch per unit time. Some thronglets might be better foragers – for example, one with higher gather rate can extract food quickly from a patch, whereas a lower rate means it eats slowly. This can determine how well a creature does in competition when multiple individuals swarm the same food source.

Gameplay & Mechanics

All these traits are genetic – meaning they are passed to offspring with slight modifications (mutations). In the game's UI, players can view the population averages of each trait at any time (e.g. "Average speed = 5.2 units, Average lifespan = 20s, …").

This is **a critical learning feature:** as the game progresses, players can see these averages shift, which is essentially observing evolution. For example, if the environment suddenly has less food, one might notice the average metabolic rate of the population decreases over generations (because high-metabolism creatures died off, leaving lower-metabolism ones reproducing more). By quantifying traits, the game gives a concrete window into the otherwise abstract concept of "allele frequencies" or trait distributions changing over time.



Offspring Inheritance: The offspring starts with trait values equal to the parent's, plus a random delta from mutation. For example, if a parent has speed 5.0, the child might have speed 5.2 or 4.7, etc. The range of possible mutation is small (perhaps ±5-10% of the trait value), so changes are gradual, reflecting real evolution more than a drastic fantasy mutation. Over multiple generations, these small changes can accumulate if there is consistent selection pressure favoring one end of a trait spectrum.- Population Initialization: At the start of a scenario, the game spawns a small founding population (say N=10 creatures) with a diversity of trait values. They might be centered around some baseline (e.g. average speed 5) but with variation. This initial genetic diversity is important so that selection has something to act upon. Players can be informed that a diverse starting population was created, reinforcing how variation is the raw material of evolution.

Through reproduction and mutation, Throng: Evolution creates a gene pool that changes over time. Players will notice that certain traits tend to drift upwards or downwards depending on what confers a survival/reproductive advantage. The game effectively visualizes microevolution – evolution on the scale of a few generations – which is exactly the level needed to teach the principle of natural selection.

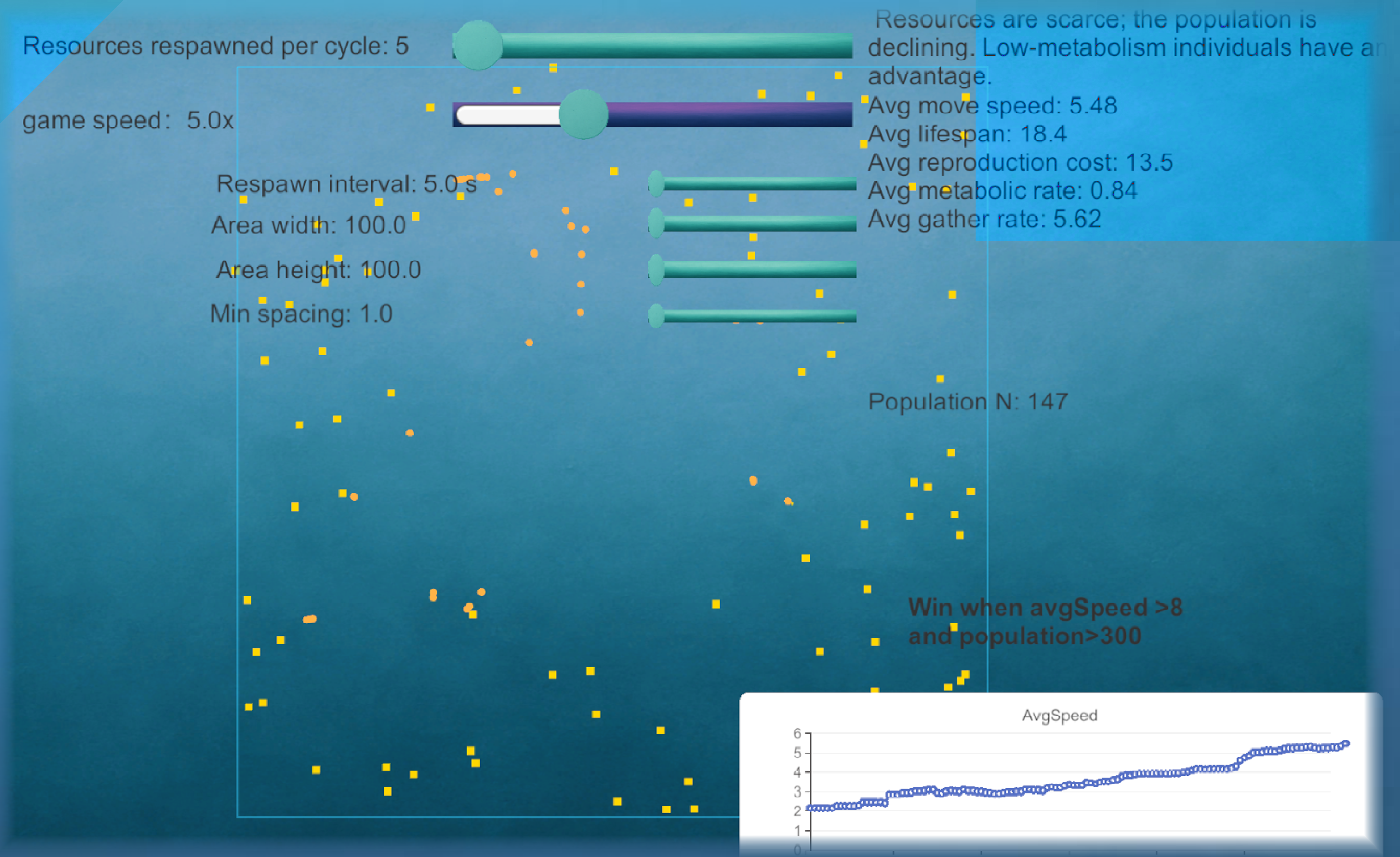
Reproduction and Mutation

Reproduction in Throng: Evolution is asexual for simplicity (each creature can reproduce on its own once it has enough energy stored). When a thronglet reproduces, it produces one offspring (a clone of itself) with some random mutations in its genes. This models the concept of genetic variation arising through mutation. Every gene (trait) of the offspring has a chance to mutate – meaning the trait value might shift slightly up or down. The mutation rates and magnitudes are tuned to ensure variability without completely randomizing the population each generation. Over many reproductions, these small variations introduce a spectrum of traits in the population.

Key points about the reproduction system:

Age and Reproduction Timing: Each creature has an internal timer or age. It may attempt reproduction after a certain maturity time and then again after each interval (provided it has accumulated the required energy). This ensures that not all creatures reproduce synchronously, and it creates overlapping generations (much like real populations).

Cost of Reproduction: As mentioned, reproducing deducts a chunk of the creature's stored energy (simulating the investment of calories into an offspring). After reproducing, the parent might be left with low energy and need to feed again before it can reproduce a second time. This can sometimes lead to death if the environment is harsh – e.g. a creature reproduces and then starves because it gave its last energy to the offspring. Such events can provoke discussion about life-history strategies (should one reproduce as soon as possible, or wait to gather more energy?).



Gameplay & Mechanics

Early testing of the game (informal observations in a classroom setting) has shown high engagement – students often react with surprise and excitement when "evolution" happens in front of them ("Look, all the fast ones died off!" followed by discussions on why that occurred). This kind of enthusiasm is exactly what is needed in science education, where student interest can sometimes be a limiting factor. While engagement is clearly boosted, we remain mindful of ensuring genuine learning takes place. The integrated explanations and the design of reflection prompts (either in-game or via an instructor's guide) aim to solidify the knowledge gained. This aligns with research indicating that games need to be complemented by debriefing or reflection to maximize learning gains[10].

In conclusion, Throng: Evolution demonstrates how a carefully crafted serious game can serve as an effective educational tool. It uses the power of play to illuminate one of science's most important ideas. By guiding a virtual species through the trials of survival, players come away with a deeper intuition for natural selection – an intuition that we hope will translate to better understanding of real-world biology and evolution.

Research References

References (selected): [3][4][1][2][8][9][11][10

[1] Misconception Monday! Survival of the Fittest, Part 1 | National Center for Science Education https://ncse.ngo/misconception-monday-survival-fittest-part-1

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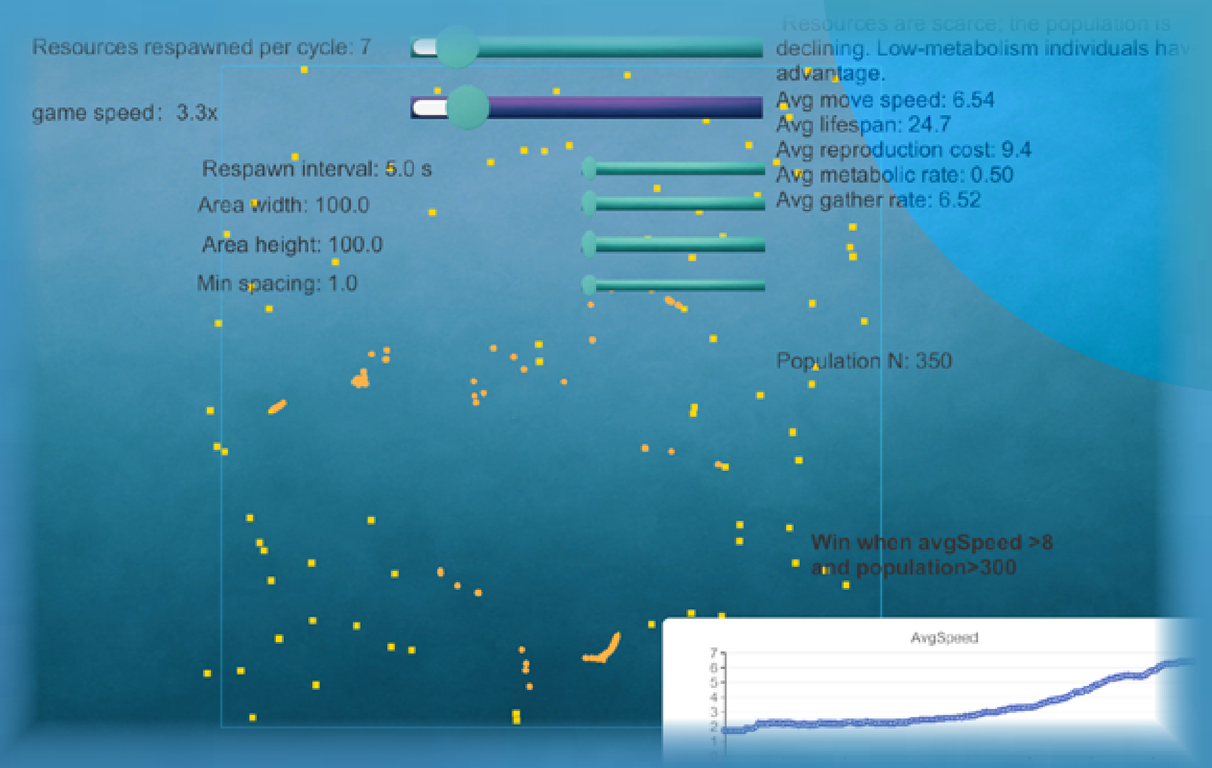
[8] [12] Three screenshots of the prey-predator simulation game | Download Scientific Diagram https://www.researchgate.net/figure/Three-screenshots-of-the-prey-predator-simulation-game_fig1_300253195

[9] [11] [14] [15] Low-tech serious games in higher education: bridging the digital divide and enhancing student thinking and performance | Humanities and Social Sciences Communications https://www.nature.com/articles/s41599-024-04341-2?error=cookies_not_supported&code=6a548d9f-c1b3-41a3-8a04-10ed6e7d789e

[10] "EFFECTIVENESS OF AN EVOLUTION-BASED SERIOUS GAME ON MOTIVATION AND ACH" by Bradford John Fruechte https://scholarworks.umt.edu/etd/12400/



UI / UX Design

Global states
Setup (modal, paused): GeneSetupUI opens on first load; Time.timeScale = 0.
Play (HUD visible): simulation running; controls and charts docked.
Pause (overlay): P key or menu; dim layer + resume / restart / quit.
Win (modal): auto-pause; "You Win!" with Next Stage.
HUD layout (16:9 reference)
Left column – Simulation Controls
Environment: Patch Respawn Count, Respawn Interval (s), Area Width / Height, Min Spacing, Hotspot Interval.
Time control: Time Scale (1–5×) with numeric readout.

Toggle: "Show Resource Area Border" (renders the spawn bounds).
Right column – Live Telemetry
Population (N) and dN/dt readouts with trend label.
K (carrying capacity) estimate, r (growth rate) estimate.
Average genes: move speed, lifespan, reproduction cost, metabolic rate, gather rate.
Teaching tip sentence (adaptive).
Bottom – Charts (XCharts)
Line charts: N, dN/dt, K, r, Avg Metabolic, Avg Speed (consistent colors & units).

Onboarding & tutorials
Scene 1 "What am I seeing?" (sprite demo)
Click-through mini-animation (thronglet moves to nearest resource → feeds → reproduces → dies).
UI highlight frame follows world objects via world → screen mapping.
Short, one-line captions with a typewriter effect (skippable on click).

Environment tutorial (in-scene, after setup)
Steps auto-generated from active sliders;
Each step:
Highlights the control (no input blocking).
Shows a concise function description (hover tooltip + inline text).
Requires one interaction to proceed (progressive disclosure).
Tooltips are component-local (UIHoverInfo) so designers author copy per slider in the Inspector.

Interaction Patterns
Sliders
Always show current value to the right (fixed-width label to avoid layout jitter).
Map thumb motion to bounded, meaningful ranges (e.g., min spacing 0.5–10.0).
Debounce heavy operations if needed (but your current listeners are light).
Time Scale slider snaps to 1.0, 2.0, 3.0, 4.0, 5.0 ticks for quick control.
Immediate feedback
Changing Area Width/Height animates the spawn border rectangle in UI (1–2 frames lerp).
Changing Respawn Count/Interval updates K estimate and N/dN/dt trend within the next sample tick, plus a brief pulse on those labels.

Charts
X-axis = simulation time (s). Y-axis autoscale per series; clamp jitter with smoothing only for display (keep raw values).
Fixed color mapping:
N (cyan), dN/dt (orange), K (violet), r (green), Avg Metabolic (magenta), Avg Speed (blue).(in the first level there's only speed)
Use the same colors in text badges for cross-referencing.
Show last 10–15 minutes in a moving window; older data decimated.